

Hyperbolic Geometry for Foundation Models: A Tutorial

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Abstract

In recent years, foundation models, such as large language models (LLMs), vision transformers (ViTs), diffusion generative models, and large multi-modal models, have revolutionized machine learning by driving significant advancements in training, inference, and deployment. However, the performance of Euclidean-based foundation models is inherently constrained by the limitations of Euclidean geometry, particularly in capturing complex structures. Recent studies have demonstrated that token embeddings often exhibit hierarchical and scale-free properties, which Euclidean spaces struggle to represent due to their linear and uniform scaling. In contrast, hyperbolic geometry, with its exponential volume growth relative to distance, offers a promising alternative for embedding tree-like structures and power-law distributions. This has led to a surge in methods incorporating hyperbolic geometry into foundation model architectures, improving the efficiency and effectiveness of token representation. To support the growing interest in hyperbolic foundation models, this tutorial provides a comprehensive and accessible review of methods, challenges, and progress in this field. We begin by introducing hyperbolic geometry and its relevance to the non-Euclidean structures in foundation models. We then explore unified approaches for hyperbolic deep learning, including key hyperbolic neural network operations and models that form the backbone of hyperbolic foundation models. Next, we delve into the technical details of hyperbolic foundation models, with a focus on hyperbolic LLMs and Transformers, vision models, and multi-modal systems. Finally, we address several key challenges and propose potential solutions, including some of our own early efforts, to pave the way for future advancements in this exciting research area.

keywords: Foundation Models; Large Language Models; Hyperbolic Geometry; Representation Learning

Tutorial Introduction

This tutorial targets machine learning researchers interested in foundation models, non-Euclidean geometry, and geometric deep learning. We explore advanced applications of hyperbolic geometry in large-scale models, extending beyond previous tutorials (Zhou et al. 2022, 2023). No differential geometry background is required, though familiarity with Riemannian manifolds is helpful.

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Core Concepts, Methods, and Modeling Frameworks.

The tutorial covers hyperbolic manifolds (Poincaré ball and hyperboloid models), Riemannian operations (exponential/logarithmic maps, parallel transport), and hyperbolic distance metrics for hierarchical data representation. Key methods include hyperbolic neural network operations (matrix multiplication, activation functions, normalization), attention mechanisms for transformers, and optimization techniques for curved spaces. Modeling frameworks span hyperbolic versions of MLPs, CNNs, RNNs, and transformers, with practical implementation strategies for scaling to foundation model sizes while maintaining numerical stability.

Anticipated Audience Size. We expect 30~50 attendees based on preliminary surveys showing high interest in academia and industry, particularly for applications in LLMs, graph analysis, image processing, drug discovery, and material analysis.

What the Audience Will Gain? Attendees will gain practical skills in implementing hyperbolic neural networks and adapting foundation models to non-Euclidean spaces. They will develop intuitive understanding of when hyperbolic geometry provides advantages and acquire the technical expertise to apply these methods to their research, enabling both immediate application and future contributions to this emerging field.

What Makes the Topic Innovative? This tutorial introduces a paradigm shift from Euclidean to hyperbolic geometry in foundation models, addressing fundamental representational limitations. While current foundation models excel in many tasks, their Euclidean embeddings struggle with hierarchical and scale-free structures in language and vision data. Hyperbolic geometry’s exponential volume growth naturally captures tree-like relationships and power-law distributions, offering mathematically principled improvements in representation efficiency. The innovation extends to practical large-scale implementation, integrating hyperbolic operations into transformer architectures, vision models, and multi-modal systems.

Illustrative Scenarios. The tutorial demonstrates concepts through hierarchical text analysis (taxonomies and knowledge graphs), social network analysis (community structures and influence hierarchies), and computer vision tasks (scene graphs and object relationships). Examples include large language models processing syntactic structures,

vision transformers handling hierarchical image features, and multi-modal systems combining textual and visual hierarchies, showing improved embedding quality and computational efficiency over Euclidean approaches.

Tutorial Content

Interactive Components

We plan on employing several strategies to encourage audience participation such as

- **Interactive Polling:** Using software such as Slido or Mentimeter, we plan on incorporating live polls to interact with the audiences.
- **Open-ended Discussions:** To further increase participation, we plan on posing open-ended questions related to our presentation to facilitate discussions.
- **Interactive Q&A:** We plan on making Q&A sessions more interactive through software such as Google Moderator.

Tutorial Outline

The tutorial will focus on hyperbolic deep learning methods for foundation models, their practical applications, and future research directions. The 3-hour tutorial is organized as follows with three 10-minute breaks strategically placed for optimal learning:

- **Introduction & Motivation** (50 minutes)
 - Overview of Euclidean foundation models and their limitations
 - Introduction to hyperbolic geometry and Riemannian manifolds
 - Motivation for incorporating hyperbolic geometry into foundation models
 - Key applications demonstrating hyperbolic advantages
- **Break** (10 minutes)
- **Hyperbolic Neural Networks as Building Blocks** (55 minutes)
 - Hyperbolic neural network operations (feature transformation, residual connections, normalization)
 - Hyperbolic neural network models (RNNs, MLPs, CNNs, GNNs)
 - Practical implementation challenges and solutions
- **Break** (10 minutes)
- **Hyperbolic Foundation Models & Applications** (60 minutes)
 - Hyperbolic Transformers and LLMs for hierarchical text processing
 - Hyperbolic vision foundation models for scene understanding
 - Hyperbolic multi-modal models for cross-domain reasoning
 - Real-world applications: knowledge graphs, social networks, recommender systems, AI for science, etc
 - Performance comparisons and efficiency gains

- **Break** (10 minutes)

- **Future Directions and Challenges** (45 minutes)

- Hyperbolic pre-trained models and scaling strategies
- Advanced hyperbolic foundation model architectures
- Open challenges: numerical stability, optimization, hardware acceleration
- Hyperbolic deep learning tools, libraries, and ecosystem development
- Emerging applications and research opportunities

Tutors

All tutors will participate **in person** for this tutorial, ensuring direct interaction with attendees, facilitating comprehensive discussions, and providing hands-on guidance during practical implementation sessions. This in-person format enables real-time collaborative problem-solving, immediate clarification of complex mathematical concepts, and personalized assistance with hyperbolic neural network implementations.

Jiahong Liu is a Ph.D. candidate at The Chinese University of Hong Kong. Her research interests center on methods and applications of hyperbolic geometry, including utilizing hyperbolic geometry to enhance foundation models and address data heterogeneity among users in domains such as recommender systems and federated learning. She has published several papers in top-tier conferences and journals, including ICDE, KDD, WebConf, AAAI, and Information Sciences. She was the area Chair of NEGEL workshop @ WWW 2025 and NeurIPS 2025.

Menglin Yang is currently an assistant professor in AI Thrust at the Hong Kong University of Science and Technology (Guangzhou), HKUST(GZ). He was previously a post-doctoral researcher at Yale University with Prof. Rex Ying and received his Ph.D. from The Chinese University of Hong Kong. His research interests include hyperbolic geometric learning and foundation models. He organized tutorials on hyperbolic graph learning at KDD 2025, KDD 2023 and ECML-PKDD 2022. He also organized a workshop on the non-Euclidean foundation model at WWW (The WebConf) 2025 and NeurIPS 2025.

Irwin King is a professor at The Chinese University of Hong Kong. His research interests generally include geometric machine learning, hyperbolic representation, data mining, and multimodal learning, with a current particular emphasis on personalization, recommendation systems, and large language models. He is IEEE fellow, ACM fellow, Associate Editor for Artificial Intelligence Journal, Neural Networks, JDSA, etc. He is the General Co-Chair of the WebConf 2020, ICONIP2020, WSDM 2011, RecSys 2013, and ACML 2015. Received multiple GRF grants on big data, machine learning, social computing, and other related projects. He was on leave with AT&T Labs Research, San Francisco to work on big data-related projects and was also teaching Social Computing and Data Mining as a Visiting Professor at UC Berkeley. He was also the host to hyperbolic realted tutors at KDD 2023 and ECML-PKDD 2022.

Societal Impact

Foundation models (Bommasani and et al. 2021), hyperbolic deep learning models (Peng et al. 2021), and the combination of the two have vast real-world and societal applications: in natural language processing (Dhingra et al. 2018; Leimeister and Wilson 2018), such as hierarchical natural language concepts capturing (Le et al. 2019), question answering systems (Tay, Tuan, and Hui 2018; Yang et al. 2024), privacy-preserving text representations (Feyisetan, Diethe, and Drake 2019), and multi-document summarization (Song, Feng, and Jing 2023); in computer vision (Mettes et al. 2024), such as visual and object detection (Ge et al. 2022; Pal et al. 2024), image segmentation (Atigh et al. 2022), and video prediction (Surís, Liu, and Vondrick 2021); in natural sciences, such as protein-folding (Villegas-Morcillo et al. 2020), single-cell-RNS-seq analysis (Ding and Regev 2021), and modeling brain activities (Shi et al. 2013); and in complex networks, such as social networks (Yang et al. 2021, 2022c, 2023) and recommender systems (Yang et al. 2022b; Chen et al. 2022; Sun et al. 2021; Yang et al. 2022a). Our tutorial would facilitate and support future research in the field of hyperbolic foundation models (He et al. 2025), which have the potential to advance the important areas listed above for wide societal impacts. Additionally, our tutorial would be widely applicable, especially in scenarios requiring large-scale models, given the critical role foundation models have had in recent years for both research and societal issues alike (Kapoor et al. 2023; Bommasani and et al. 2021).

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Supplementary Materials: Revelant Works from Tutors

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